LONGMAN GCSE BIOLOGY WORKSHEETS

6: Mendel's peas

This worksheet tests your understanding of Mendel's breeding experiments and looks his results in light of the discovery of genes.

Gregor Mendel published the results of his experiments on the genetics of the garden pea plant in 1865. He carried out a number of experiments on contrasting pairs of phenotypic characteristics, such as green or yellow pod colour. For example, when he crossed plants that were pure-breeding for green pods with plants that were pure-breeding for yellow pods, he found that all the offspring (called the F_1 generation) had green pods. However, when he crossed-fertilised these F_1 plants, the next generation (called the F_2) consisted of some plants that had green, and others yellow, pods. He found similar results for several other contrasting pairs of characteristics.

The F_2 offspring for the pod colour experiment and six other experiments on different contrasting characters is shown in the table:

Characters of 'parent' plants	**F_1 generation**	**Numbers in F_2 generation**	**Ratio of numbers in F_2**
green pod × yellow pod	all green	428 green : 152 yellow	2.82 : 1
inflated pod × constricted pod*	all inflated	882 inflated : 299 constricted	2.95 : 1
round peas × wrinkled peas	all round	5474 round : 1850 wrinkled	
yellow peas × green peas	all yellow	6022 yellow : 2001 green	
long stem × short stem	all long	787 long : 277 short	
purple flower × white flower	all purple	705 purple : 224 white	
flower along stem × flower at tip	all along stem	651 along stem : 207 at tip	

* Inflated pods look fat – constricted pods have the peas visible in them.

Questions

1. Why could Mendel not have used the word 'genetics'?

2. What do we mean by the word 'phenotype', as in 'phenotypic characters'?

3. Calculate the missing ratios of numbers in the F_2 generation to complete the last column of the table.

4. Using the symbol 'R' for the allele for round peas and 'r' for the allele for wrinkled, draw genetic diagrams to explain the results of the cross between plants with round and wrinkled peas, for both the F_1 and F_2 results. Use these words in the correct places on your diagram **homozygous**, **heterozygous**, **dominant**, **recessive** and **genotype**.

5. What word would we now use to describe the genetics of the 'pure-breeding' parents?

6. Your answer to question 4 should predict that the phenotypes of the F_2 generation are in the ratio 3 : 1. The ratios in the table are very close to, but not exactly equal to, 3 : 1. There are a number of reasons for this difference. Suggest three.

When Mendel published his findings in 1865, no one knew about the existence of chromosomes, so Mendel could not explain his findings in terms of structures in the cell. In the light of this handicap, his work was an outstanding achievement, because it led him to propose that

- *'Particles of inheritance'* (which we now called genes) *exist*
- *Particles are present in pairs in 'body' cells*

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- *Pairs of particles separate during the formation of gametes, so that each gamete only carries one of each pair*
- *Particles can have dominant and recessive forms (alleles).*

Nowadays we know about chromosomes and meiosis, so we can explain what is happening inside the cell to produce Mendel's results.

7. By referring your genetic diagrams for crosses between round and wrinkled peas (question 4), draw a series of simple diagrams of cells (containing just one pair of chromosomes for simplicity) to show what happens to the alleles for round and wrinkled phenotypes during formation of gametes (meiosis) and then during fertilisation.